Monthly Intelligence Report

Edition 001 — The Tyrol-Vorarlberg Alpine Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

1,406

1,144 HV · 262 MV

GRID LINES MAPPED

11,267

~15,000 km coverage

MC SIMULATIONS

14.06M

10,000 per substation

PUBLIC DATA SOURCES

35

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 346,000+ source data points per run, executes 14.06 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and produces full uncertainty quantification across 1,406 substations spanning 85 Bezirke (administrative districts) and 11,267 grid lines covering ~15,000 km.

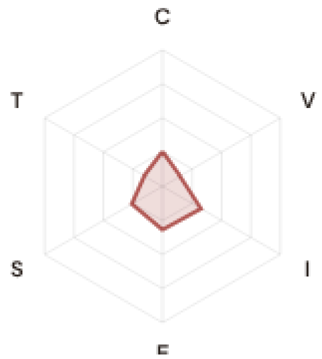
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 35 verified public sources, making the methodology fully auditable and reproducible. The Index captures Austria's unique challenges: Alpine infrastructure exposure, hydropower dependency (60%+ of renewable generation), east-west transmission constraints, and cross-border interconnection with Germany, Switzerland, Italy, and the Czech Republic. This inaugural Austrian edition prioritises the critical Tyrol-Vorarlberg Alpine Corridor, where APG transmission bottlenecks intersect with the highest hydropower concentration in Central Europe.

Substation in Focus

Umspannwerk Merching

Reutte, Tirol · 110 kV

0.252	Medium	0.086	39th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.252 / 0.30 (84%)
- **I** Infrastructure: 0.325 / 0.25 (130%)
- **S** Saturation: 0.259 / 0.20 (130%)
- **V** Voltage: 0.148 / 0.10 (148%)
- **E** Economic: 0.317 / 0.10 (317%)
- **T** Transition: 0.153 / 0.05 (307%)

MODIFIERS

- **R3 Consequence:** 0.928 ↓ dampens
- **R4 Graph Criticality:** 0.967 ↓ dampens
- **R6a Restoration:** 1.052 ↑ amplifies
- **R6b Network Topology:** 1.000 → neutral
- **R7 Digital Readiness:** 1.014 ↑ amplifies

This month's spotlight — automatically selected for April 2026 — is **Umspannwerk Merching** in Reutte, Tirol. With an R_final of 0.252 (Medium band, 39th fleet percentile), its risk profile is dominated by the E (Economic) component at 317% of its weight ceiling, followed by T (Transition) at 307%. Modifiers collectively shift R_base by 4.0%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Austria's Energiewende transition planning and grid modernisation initiative require.

CYBER HIGH EXPOSURE

416

29.6% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0013

Range [0.99, 1.05]

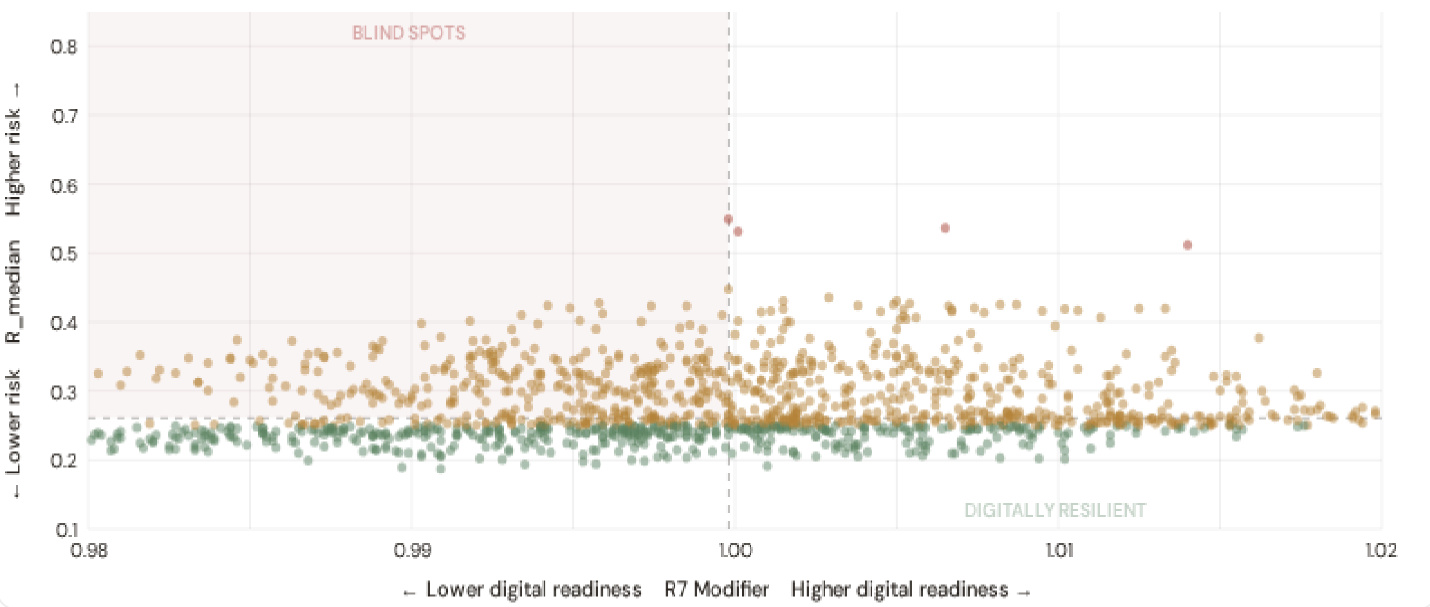
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9998). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in . Across the full fleet, 416 substations (29.6%) carry a HIGH cyber-exposure classification, while 420 (29.9%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

354 substations (25.2%) High/Critical: **0** (0.0%) Avg R: **0.268** Avg E: **0.303** / 0.10

Low 182 Med 172 High 0 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

349 substations (24.8%) High/Critical: **2** (0.6%) Avg R: **0.283** Avg E: **0.303** / 0.10

Low 99 Med 248 High 2 Crit 0

🏢 Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

354 substations (25.2%) High/Critical: **2** (0.6%) Avg R: **0.284** Avg E: **0.303** / 0.10

Low 105 Med 247 High 2 Crit 0

🌾 Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

349 substations (24.8%) High/Critical: **3** (0.9%) Avg R: **0.294** Avg E: **0.304** / 0.10

Low 102 Med 244 High 3 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate for Austria places average VoLL at €11.2/kWh for industrial customers and €3.0/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Tirol (105), Salzburg (48), Vorarlberg (40)** — regions where industrial corridors depend on uninterrupted power for continuous processes (paper, chemicals, food). A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. Austria's distributed economic fabric — strong across all Bundesländer — means the total economic exposure is spread more evenly than in less balanced economies. This is a structural advantage, but it does not eliminate the urgency of identifying and protecting the highest-consequence substations. This cross-referencing of economic fabric with grid condition is unique to the SSI.

B.3 Bezirk Digital Readiness Ranking

Bezirke (districts) ranked by average R7 modifier — lower values indicate weaker digital infrastructure. Cross-referenced with average R_median to identify Bezirke combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 BEZIRKE

Longer bar = higher R7 = better digital infrastructure.

1	Vorarlberg	0.9969
2	Salzburg	0.9973
3	Wien	0.9992
4	Burgenland ⚠ high R	1.0002
5	Oberösterreich ⚠ high R	1.0006
6	Kärnten ⚠ high R	1.0007
7	Tirol	1.0026
8	Niederösterreich ⚠ high R	1.0031
9	Steiermark ⚠ high R	1.0047

Regional analysis reveals digital infrastructure variation across Austria's 9 Bundesländer. **Vorarlberg** has the lowest average R7 (0.9969), while **Steiermark** leads at 1.0047 — a spread of 0.0079. **0 Bundesländer** combine below-median digital readiness with above-median grid risk, making them priority targets for E-Control-coordinated modernisation investment.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor across the Austrian fleet. Key reference points: fleet average R7 = 1.0013, median = 0.9998; 416 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging Bundesländer where digitalisation investments translate into measurable R7 shifts.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. Austrian sources include: E-Control monitoring reports, Statistik Austria STATPOP, APG Netzentwicklungsplan, ZAMG/GeoSphere climate data, and BEV geographic data. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
35 95 variables	**5** High-frequency feeds	**22** Regular refresh cycle

STATIC / DERIVED

8

Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
APG	APG Transmission System Operator	MONTHLY	Control area	3 vars
APG-N	APG Netzentwicklungsplan	MONTHLY	TSO zone	2 vars
BMI-AT	BMI Civil Protection Austria	CONTINUOUS	Bezirk	1 var
STAT-KFZ	Statistik Austria KFZ Registry	QUARTERLY	Bezirk	1 var
FFG	FFG Innovation Agency Austria	QUARTERLY	Bezirk	1 var
ECONT	E-Control Austria	ANNUAL	Bezirk	8 vars
ECREG	E-Control Grid Registry	ANNUAL	Bezirk	5 vars
STAT	Statistik Austria	ANNUAL	Bezirk (GKZ)	8 vars
UBA-AT	Umweltbundesamt Austria	ANNUAL	Bezirk / station	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
BMNT	BMNT Sustainability & Tourism	ANNUAL	Bundesland	2 vars
BMLRT	BMLRT Regional Development	ANNUAL	Bundesland	1 var
BMF-AT	BMF Austria Fiscal Statistics	ANNUAL	Bezirk	1 var
BEV	BEV Federal Office of Metrology	ANNUAL	Bundesland	1 var
WIFO	WIFO Economic Research	ANNUAL	Regional	2 vars
OEE	Oesterreichs Energie (Association)	ANNUAL	Bundesland	2 vars
OEROK	ÖROK Regional Indicators	ANNUAL	Bezirk	2 vars
ECMON	E-Control Monitoring Report	ANNUAL	DSO-level	2 vars
OROK-S	ÖROK Spatial Planning	ANNUAL	Bezirk	1 var
IHS	IHS Institute for Advanced Studies	ANNUAL	Bundesland	1 var
AEA	Austrian Energy Agency	ANNUAL	Bundesland	1 var
GBA	GBA Geological Survey Austria	STATIC	Bezirk	3 vars

D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
GKZ	Gemeindekennziffer / GKZ Registry	STATIC	Bezirk (5-digit)	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	Bezirk	1 var
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
ZAMG	GeoSphere Austria (ex-ZAMG)	DAILY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (3 sources) — Grid load + procurement data

E-Control Registry · APG · APG Netzentwicklungsplan

Quarterly (2 sources) — DER registry + business data + macro indicators

Statistik Austria KFZ · FFG

Annual (20 sources) — Core grid quality + demographics + environment

E-Control · Statistik Austria · WIFO · OeE · BMNT · BMLRT · BMF-AT · UBA-AT · EEA · Eurostat · DESI · ÖROK · E-Control Monitoring · ÖROK Spatial · IHS · AEA · BEV · GeoSphere · JRC · BMI-AT

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · GBA · ISO 9223 · GKZ/AGS · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE
1,406
substations scored

MEDIAN R
0.261
P5=0.216 · P95=0.397

HIGH + CRITICAL
7
0.5% of fleet

HIGH CONFIDENCE
100%

1406 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 488 (34.7%) Medium 911 (64.8%) High 7 (0.5%) Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **1,406 substation scores remain unchanged** this edition. The fleet median R of 0.261 (mean 0.282) reflects a distribution skewed toward the lower-to-medium bands, with 34.7% in Low and 64.8% in Medium. The first material score changes are expected when E-Control publishes the updated TIQE tables (affecting C1–C4 and R6a) and when APG refreshes the DER registry (affecting T1).

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Bezirk-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — BMF + ÖROK + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — Statistik Austria net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	E-Control Registry upgraded to bulk download — Bezirk-level DER registry	§12
G2	DATA	GBA upgraded to WMS live API — Bezirk-level geological data	§12
G3	DATA	OSM upgraded to Overpass API — 1,406 real substations	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — APG Netzentwicklungsplan, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Tyrol-Vorarlberg Alpine Corridor: Transmission Bottlenecks Meet Hydropower Density

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Tyrol-Vorarlberg corridor · **002** Oberösterreich (Industrial grid, VOEST) · **003** Wien (Urban density, Wiener Netze) · **004** Niederösterreich (Wind integration, largest Bundesland) · **005** Steiermark (Green hydrogen, industrial transition) · **006** Kärnten (Southern Alpine, cross-border Italy) · **007** Salzburg (Hydropower hub, tourism load) · **008** Burgenland (Wind champion, Pannonian corridor) · **009** Vorarlberg (Isolated west, CH/DE cross-border) · **010** Tirol (Brenner transit, ski tourism seasonality) · **011** Annual Review · **012** Methodology Deep-Dive.

D.1 Why the Alpine Corridor, Why Now

CORRIDOR SUBSTATIONS

593

126 HV · 467 MV

HIGH BAND

0

0.0% of corridor

CORRIDOR MEDIAN R

0.248

Fleet median: 0.261

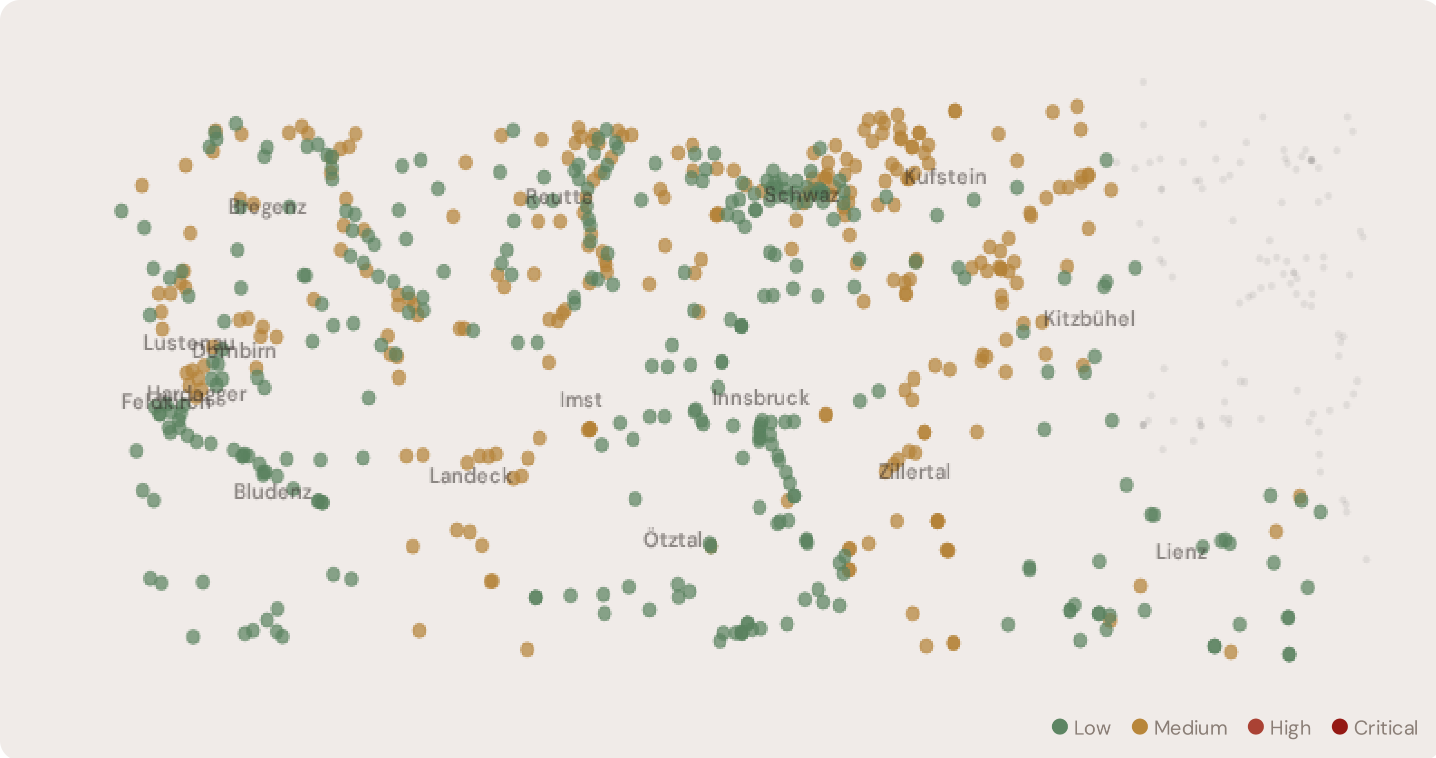
WORST PROVINCE

Landeck

Avg R: 0.338 · 18 substations

The Tyrol-Vorarlberg Alpine Corridor hosts **593 substations**, concentrated in Austria's two westernmost Bundesländer. This region presents a unique combination of engineering and economic challenges: Alpine terrain creates infrastructure vulnerability, hydropower generates 60%+ of Austria's renewable supply but introduces seasonal volatility, and APG's transmission network faces critical bottlenecks (380 kV Salzburgleitung, Arlberg) limiting east-west power flow and forcing reliance on transit through Germany and Switzerland. Risk profile is concentrated: **0 substations (0.0%)** are classified High, with only **324** in the Low band. 19% of corridor substations score above the national fleet median of 0.261. The corridor median R of 0.248 places Tirol-Vorarlberg among Austria's most stressed transmission corridors.

D.2 Corridor Map and Risk Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Malga Mare	Landeck	0.356	Medium	0.257	0.145	0.314	0.292	0.252	0.147	—
Zillertal Junction-108	Zillertal	0.355	Medium	0.268	0.158	0.350	0.329	0.242	0.149	—
Umspannwerk Nauders	Landeck	0.353	Medium	0.264	0.148	0.322	0.309	0.259	0.148	—
Glorenza	Landeck	0.352	Medium	0.256	0.157	0.323	0.313	0.264	0.148	—
Tiwag Umspannwerk Ötztal	Imst	0.348	Medium	0.260	0.154	0.327	0.299	0.247	0.146	—
APG Umspannwerk Westtirol	Imst	0.348	Medium	0.251	0.157	0.335	0.295	0.252	0.156	—
Landeck Junction-125	Landeck	0.348	Medium	0.250	0.157	0.329	0.311	0.241	0.144	—
Lappach - Lappago	Zillertal	0.347	Medium	0.261	0.151	0.333	0.313	0.268	0.155	—
UW Tobadill	Landeck	0.347	Medium	0.241	0.151	0.312	0.300	0.265	0.145	—
Zillertal Junction-124	Zillertal	0.346	Medium	0.272	0.148	0.345	0.308	0.253	0.135	—

… 583 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — ALPINE VS FLEET

C — Continuity	0.221 vs 0.331 (-33%)
I — Infrastructure	0.320 vs 0.320 (0%)
S — Saturation	0.257 vs 0.257 (0%)
V — Voltage	0.150 vs 0.150 (-0%)
E — Economic	0.303 vs 0.303 (-0%)
T — Transition	0.150 vs 0.150 (-0%)

MODIFIER AVERAGES — ALPINE VS FLEET

R3 Consequence	0.954	fleet 0.957 ↓
R4 Graph Crit.	1.021	fleet 1.010 ↑
R6a Restoration	1.039	fleet 1.032 ↑
R6b Seismic	1.000	fleet 1.000 →
R7 Digital Read.	1.001	fleet 1.001 →

The dominant risk driver across the Alpine Corridor is **E (Economic)**, averaging **0.303** against a fleet average of 0.303 — occupying **303% of its weight ceiling**. The second contributor is **T (Transition)** at **0.150** vs fleet 0.150. Among modifiers, the **R6b seismic modifier** averages **1.000**, capturing the Alpine region's moderate seismic exposure — a unique risk dimension captured by the SSI. The **Transition component (T)** at **0.150** reflects the accelerating DER penetration and renewable integration driving portfolio stress in these corridors.

D.4 Province Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Landeck	18 subs · avg R = 0.338 · H:0 M:18
Imst	6 subs · avg R = 0.335 · H:0 M:6
Zillertal	30 subs · avg R = 0.332 · H:0 M:30
Lustenau	7 subs · avg R = 0.268 · H:0 M:7
Kufstein	103 subs · avg R = 0.258 · H:0 M:88
Reutte	116 subs · avg R = 0.250 · H:0 M:61
Bregenz	77 subs · avg R = 0.249 · H:0 M:36
Schwaz	49 subs · avg R = 0.245 · H:0 M:14
Kitzbühel	9 subs · avg R = 0.243 · H:0 M:1
Lienz	37 subs · avg R = 0.241 · H:0 M:5
Ötztal	29 subs · avg R = 0.237 · H:0 M:1
Innsbruck	59 subs · avg R = 0.236 · H:0 M:2

Hardegger 2 subs · avg R = 0.221 · H:0 M:0

Bludenz 31 subs · avg R = 0.220 · H:0 M:0

Feldkirch 15 subs · avg R = 0.205 · H:0 M:0

Dornbirn 5 subs · avg R = 0.195 · H:0 M:0

The province-level breakdown reveals concentration of risk. **Landeck** leads with a mean R of 0.338 across 18 substations, while **Dornbirn** scores 0.195 — a spread of 0.143. APG's transmission network focuses on interconnection and east-west power transfer; at this level, asset-by-asset granularity is essential for investment decisions. The spatial pattern identifies the specific bottleneck substations constraining the Brenner transit corridor and the Arlberg transmission link.

D.5 Implications

0 Alpine Corridor substations score R ≥ 0.650, in the upper quartile of national risk. For Austrian grid modernisation priorities, this analysis identifies **Landeck** as the highest-priority target — where transmission constraints are acute, hydropower integration is intensive, and cross-border flows are heaviest. The SSI's uniqueness is that it connects engineering condition (continuity, infrastructure age) with network topology (R4) and geographic exposure (Alpine seismic, climate degradation trends). This is the evidence base for APG's €2.5B+ transmission investment programme.

SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card to position Austria within the ENTSO-E and ACER benchmarking framework. This edition: **Austria as a Transit Hub** — because the Tyrol-Vorarlberg corridor analysis reveals Austria's critical role in European power flows. Future editions rotate through: Hydropower vs Nordic peers, Cross-border pricing, Renewable penetration vs. EU average, Grid digitalisation benchmarking (DESI), etc. The underlying data updates annually with ACER reporting cycles.

AUSTRIAN SUBSTATIONS
1,406
SSI v4.0.2 coverage

CROSS-BORDER FLOWS
~70%
Transit power through Austria

HYDROPOWER SHARE
60%+
Renewable generation

GRID OPERATORS
8
DSOs + 1 TSO (APG)

Austria's position in Central Europe: Austria sits at the intersection of four major European grid zones — German Übertragungsnetz to the north, Italian Terna to the south, Swiss Swissgrid to the west, and Czech operators to the east. APG (Austrian Power Grid) manages the 380/220/110 kV transmission network spanning 10,600 km across all 9 Bundesländer. The 2023 ACER grid adequacy assessment rated Austria as "adequate" but flagged transmission constraints in the Tyrol-Vorarlberg Alpine region as the single largest bottleneck limiting Renewable-to-load matching during high-wind periods. The SSI Index provides the substation-level intelligence that Austria's Energiewende planning requires — identifying not regional aggregate problems, but specific assets that demand urgent action.

E.1 Austria in European Benchmarking

RENEWABLE PENETRATION

Austria: 76% renewable electricity (2023) — among highest in EU. Driven by hydropower (60%) + wind (18%) + solar (4%). This creates unique grid stress: seasonal hydropower variation (dry summers) combined with wind volatility (intermittent Alpine flows) creates a transmission system under constant pressure. The ENTSO-E 2023 Winter Outlook identified Austria as a "critical node" in European power flows.

DIGITALISATION (DESI)

Austria: Rank 8/27 EU on Digital Economy & Society Index (2023). E-Control's mandatory smart meter rollout (65% coverage) and APG's digitalisation programme are advancing, but rural Alpine areas lag. The SSI's R7 cyber-readiness modifier captures this gradient — identifying DSOs combining high grid stress with below-average digital infrastructure, prioritising investment targets for E-Control-funded modernisation.

E.2 Cross-Border Interconnection

Austria is the geographic linchpin between North-South Europe (Germany ↔ Italy transit) and East-West (Germany ↔ Czech ↔ Hungary). Approximately 70% of power transmitted through the Austrian grid originates from other countries. This creates a structural vulnerability: any outage at a critical juncture (e.g., Brenner corridor, Arlberg transmission) cascades across European price zones. The SSI Index's R4 (Graph Criticality) modifier quantifies this precisely — identifying which Austrian substations are highest-impact on the European grid topology.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DSO and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how SSI Index data feeds directly into investment-grade analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage
SSI v4.0 component taxonomy, normalisation, MC engine	DNO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration

LAYER 2 §0.4.1 · Edition 001 · April 2026

DNO Value Stack (D1–D5)

What a BESS is worth to the distribution network operator

The DNO value stack quantifies five categories of value that a battery creates for the DSO at each substation. D1 (Avoided Outage Cost) uses the SSI's C component and Austrian SAIDI data. D2 (Deferred Reinforcement) leverages S and I components to calculate how BESS deployment postpones upgrades. D3–D5 cover reactive power support, loss reduction, and fault level management. The Austrian regulatory framework (E-Control tariff regulation, ACER incentive mechanisms) directly shapes the valuation. Each component maps to specific economic benefits — the Index doesn't just measure risk, it prices the business case for intervention.

KEY CONCEPTS

- → D1: Avoided outage from C component + SAIDI
- → D2: Deferred reinforcement from S, I components
- → Austrian DSOs: Wiener Netze, TINETZ, VKW Netz, etc.
- → E-Control regulated tariff framework

LIVE SSI DATA CONNECTION

Across the 1,406-substation Austrian fleet, 7 substations (0.5%) are classified High or Critical — these are the priority candidates for BESS deployment where DNO and community value stacks are largest.

Coming next month: LAYER 1 Data Architecture — Component Taxonomy — *How the SSI's 6 components and 20 sub-metrics are sourced and validated*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

Carinthian Cross-Border Flows

Deep-dive into Kärnten's grid risk profile — substation map, component analysis, province breakdown. European Context Card: undefined.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

Statistik Austria KFZ, FFG are due for refresh in Q3. Impact: DER registry updates (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **E-Control Austria** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Kärnten**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Austrian utility, E-Control, Stadtwerk, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 1,406 substations · 11,267 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5
